Summarizing Clinical Text Using NLP Models

Isaac Adebayo[1], Charnisha Azubuike[1], and Michella Maddox-McGhee[1]

[1]Meharry Medical College

November 2025

Abstract

Clinical notes and medical abstracts contain a lot of important information, but they are often long, repetitive, and difficult to review quickly. This project explores how natural language processing (NLP) can help shorten clinical text while still keeping the main meaning. We used a medical abstract dataset and tested four different models: SciFive, Pegasus, and BioGPT for abstractive summarization, and ClinicalBERT for extractive summarization. Before applying the models, we cleaned the text, tokenized it, built n-grams and Word2Vec embeddings, and split the text into smaller chunks that the models could handle. Each model produced its own version of the summary, and we compared the results using ROUGE, BLEU, and BERTScore. These metrics helped us see how much information the summaries kept from the original text. The overall goal was to understand which type of summarization—abstractive or extractive—works better for clinical-style writing. Our findings highlight differences between the models and give early insight into what approaches might be useful for summarizing patient records in real clinical settings. The result derived from metrics analysis indicates that BioGPT outperformed the other model compared based on its BLEUScore, precision, accuracy, and F1-score on all evaluation metrics. The result indicates a preference for abstractive summarization over extractive summarization when it comes to summarizing clinical records.

1 Introduction

Electronic health records have made it easier to store patient information, but they also bring a new challenge: there is simply a lot of text to read. (Dahl et al., 2025) Providers often deal with long clinical notes that contain detailed descriptions of symptoms, history, assessments, and treatment plans. While this information is valuable, it can take time to go through everything, especially in busy settings where decisions need to be made quickly. Because of this, there is growing interest in using natural language processing (NLP) to help summarize clinical text automatically. (Van Veen et al., 2023)

1.1 Motivation

The problem matters because long notes slow down clinical workflows, increase cognitive load, and can lead to important information being overlooked. (Jerfy, Selden, & Balkrishnan, 2024) A good summary could make it easier for providers to review key points without digging through pages of text.

1.2 Problem Statement

The specific problem our project addresses is how to automatically generate a shorter, meaningful summary of clinical-style text in a way that keeps the essential medical information intact.

1.3 Research Question

Which NLP summarization approach abstractive or extractive produces more accurate and useful summaries for clinical text?

1.4 Background and Context

Summarization models generally fall into two categories:

- **Abstractive summarization:** creates new sentences that capture the meaning of the text.
- **Extractive summarization:** selects the most important sentences directly from the original document.

Clinical text is challenging for both methods because it includes domain-specific vocabulary, abbreviations, and long descriptive passages.

1.5 Contributions

Our project contributes the following:

- We compare three abstractive models (SciFive, Pegasus, BioGPT) with one extractive model (ClinicalBERT).
- We apply a full preprocessing pipeline, including cleaning, tokenization, n-grams, Word2Vec embeddings, and chunking.
- We evaluate model outputs using ROUGE, BLEU, and BERTScore.
- We highlight strengths, weaknesses, and practical differences between extractive and abstractive summarization in a clinical context.

1.6 Paper Structure

The rest of the paper is organized as follows: Section 2 reviews related work in clinical NLP summarization. Section 3 describes our methodology, including the dataset, preprocessing steps, and model setup. Section 4 presents the results, and Section 5 discusses the findings, limitations, and future directions.

2 Related Work

Research on summarizing medical text has increased over the past few years, mainly because clinical notes take a long time to read and often contain more information than providers need at one time. Earlier work in this area focused on extractive summarization. These methods usually relied on identifying key sentences based on relevance, term frequency, or sentence importance. Models like ClinicalBERT and Bio_ClinicalBERT became popular because they were trained on biomedical or clinical text, which helped them understand medical terminology better than general-purpose models. Extractive methods tend to be reliable at keeping the original meaning, but the summaries can still feel long or not very readable.

As transformer models grew more common, researchers began shifting toward abstractive summarization. This approach is more flexible because it allows the model to rewrite the text. Pegasus is one example that has been used widely in summarization tasks, although it was originally trained on news datasets. SciFive is another abstractive model built specifically for biomedical and clinical use, making it more suited for handling medical terminology. BioGPT has also been used for generating and summarizing biomedical text and has been shown to produce smoother, more natural-sounding sentences. Additionally, Neupane et al. (2024) introduced a framework that uses large language models to create patient-centric medical responses, demonstrating the growing role of LLMs in clinical text processing.

Even though these models show promise, the results across studies are not consistent. Some papers report that abstractive models produce more readable summaries, while others find that extractive models keep more accurate details from the original text. Many studies point out common issues: limited context windows, the difficulty of representing detailed medical concepts, or the lack of large, high-quality training datasets for the clinical domain.

2.1 Gap Our Work Fills

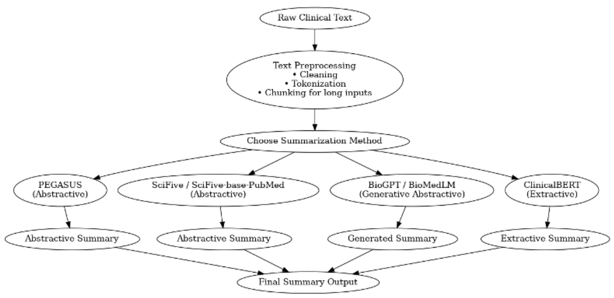
Although several models have been tested individually, fewer studies directly compare multiple abstractive and extractive models on the same clinical-style dataset, especially using a simple, controlled workflow like ours. Our project fills this gap by running all four models under the same preprocessing steps and evaluating them with the same metrics. This helps give a clearer, side-by-side understanding of how these approaches behave on clinical-type text.

3 Methodology and Approach

The goal of this project is to summarize clinical text using different machine learning models to determine which model performs best. The summarization task is divided into two categories: **abstractive summarization** and **extractive summarization**. Abstractive summarization rewrites text in a concise manner using new or original words, while extractive summarization selects key sentences directly from the original text.

The summarization process is aligned with the SOAP (Subjective, Objective, Assessment, Plan) structure commonly used in clinical documentation. The intended audience for the summarized text is medical clinics.

3.1 Flowchart



3.2 Dataset Information

The dataset was obtained from Kaggle and includes both a training and test set. Only the test dataset was used in this project to avoid training bias and to better evaluate the strength of each model. The medical abstract dataset includes five class labels: neoplasm, digestive system diseases, nervous system diseases, cardiovascular diseases, and general pathological conditions. The dataset is approximately 18 MB and the analysis was performed in Google Colab. Due to memory limitations, a text sample of six cases was used.

3.3 Preprocessing Steps

The preprocessing stage included:

- Cleaning text using regular expressions (e.g., removing extra spaces with `re.sub`).
- Converting all text to lowercase to improve model performance.
- Tokenizing the cleaned text using a word-level tokenizer, which splits the clinical text into individual tokens.

3.4 Feature Engineering

Two feature engineering techniques were used:

N-grams: Bigrams and trigrams were generated to identify recurring phrases and predict word sequences. Each produced a sample of twenty outputs.

Word2Vec: Word2Vec was used to generate word embeddings, capturing semantic relationships between words. Hyperparameters included a window size of 7, 20 training epochs, and a vector size of 150.

3.5 Chunking

Because most summarization models have a maximum token limit (commonly 1024 tokens), the cleaned text sample was divided into chunks of approximately 450 words. This reduced computational load in Google Colab and ensured compatibility with each model.

3.6 Models Used

Four different models were applied:

3.6.1 SciFive (Abstractive)

- Transformer: `AutoModelForSeq2SeqLM`
- Model: `razent/SciFive-base-PubMed`
- Hyperparameters: `num_beams=4`, `early_stopping=True`, `no_repeat_ngram_size=3`, `repetition_penalty=1.2`, `max_len=1024`, `min_len=40`

3.6.2 Pegasus (Abstractive)

- Transformers: `PegasusTokenizer`, `PegasusForConditionalGeneration`
- Model: `google/pegasus-cnn_dailymail`
- Hyperparameters: `num_beams=4`, `early_stopping=True`,

`no_repeat_ngram_size=3`, `repetition_penalty=1.2`, `max_len=1024`, `min_len=40`

3.6.3 BioGPT (Generative Abstractive)

- Transformer: `AutoModelForCausalLM`
- Model: `microsoft/BioGPT`
- Hyperparameters: `num_beams=4`, `early_stopping=True`, `no_repeat_ngram_size=3`, `repetition_penalty=1.2`, `max_new_tokens=160`

3.6.4 ClinicalBERT (Extractive)

- Transformer: `SentenceTransformer`
- Model: `emilyalsentzer/Bio_ClinicalBERT`
- Method: cosine similarity between sentence embeddings

3.7 Abstractive model hyperparameters explained

- Early stopping was set to **"True"** to prevent overfitting
- no_repeat_ngram_size had a size of **"3"** to prevent the model from getting stuck in a loop
- repetiton_penalty was set to **"1.2"** to punish the model for repeating certain word
- max_len_token was set to **"1024"** which is the maximize the amount of words processed
- min_len_token was set to **"40"** which is the minimum amount of words required

3.8 Extractive model Equation

The cosine similarity formula used is:

$$\text{Cosine Similarity}(A, B) = \frac{A \cdot B}{\|A\| \|B\|}.$$

Table 1: **A description of each model used to summarize the clinical medical text**

Model	Model Name	Transformer	Hyperparameter	Type
SciFive	razent/SciFive-base-PubMed	AutoModelForSeq2SeqLM	num beam search early stopping repetition penalty no repeat ngram	Abstractive
Pegasus	google/pegasus-cnn_dailymail	PegasusTokenizer and PegasusForConditional-Generation	num beam search early stopping repetition penalty no repeat ngram	Abstractive
BioGPT	microsoft/BioGPT	AutoModelForCausalLM	num beam search early stopping repetition penalty no repeat ngram	Abstractive
BERT	emilyalsentzer/Bio_ClinicalBERT	Sentence Transformer	num beam search early stopping repetition penalty no repeat ngram	Extractive

4 Experiments & Results

4.1 ROUGE Scores

The results of the analysis clearly indicated that BioGPT outperformed all other models across the evaluated metrics. Examination of the ROUGE scores (including ROUGE 1, ROUGE 2, and ROUGE 3) showed that BioGPT consistently achieved the highest F1-scores. This pattern suggested that BioGPT was more effective at capturing essential information from the reference summaries while also maintaining accuracy in the words and phrases it generated.

Although PEGASUS, SciFive, and ClinicalBERT demonstrated some level of alignment with the reference texts, their overall performance remained notably lower, particularly in terms of balanced precision and recall.

Table 2: **ROUGE evaluation metrics (Precision, Recall, and F1-score) for all summarization models.**

Model	Metric	Precision	Recall	F1
BioGPT	ROUGE-1	0.9927	0.3382	0.5046
	ROUGE-2	0.9782	0.3327	0.4966
	ROUGE-L	0.9879	0.3366	0.5021
PEGASUS	ROUGE-1	0.9210	0.0288	0.0558
	ROUGE-2	0.8108	0.0247	0.0480
	ROUGE-L	0.8421	0.0263	0.0511
SciFive	ROUGE-1	0.8281	0.0436	0.0828
	ROUGE-2	0.3015	0.0157	0.0298
	ROUGE-L	0.4218	0.0222	0.0422
ClinicalBERT	ROUGE-1	1.0000	0.0897	0.1647
	ROUGE-2	0.9722	0.0865	0.1589
	ROUGE-L	1.0000	0.0897	0.1647

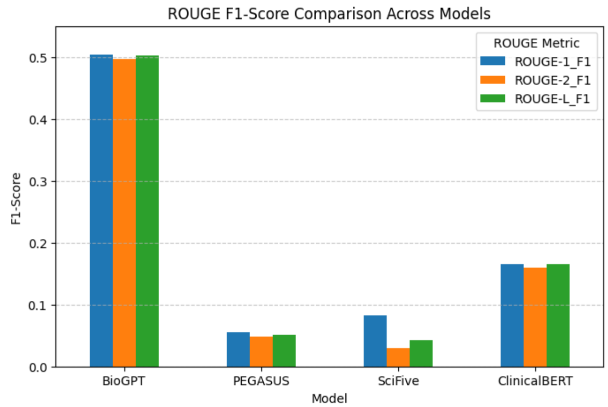


Figure 1: ROUGE F1-Score across all 4 models

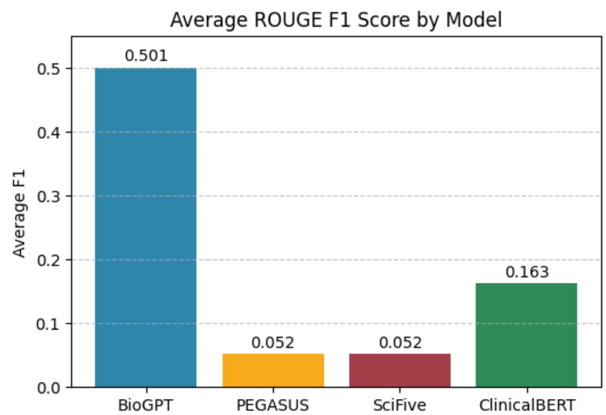


Figure 2: Average ROUGE F1 Score for all 4 models combined

4.2 BLEU Scores

The BLEU evaluation further supported these findings. BioGPT achieved a BLEU score of approximately 14, which was substantially higher than the scores produced by PEGASUS, SciFive, and ClinicalBERT. Because BLEU measures how closely the model's phrasing matches the reference summary, the higher score reflected BioGPT's stronger ability to produce outputs that aligned with the original writing. In contrast, the lower BLEU scores of the competing models indicated reduced consistency and a greater tendency to diverge from the reference summaries.

Model	Metric	BLEU Score
BioGPT	BLEU	14.1886
PEGASUS	BLEU	4.0977
SciFive	BLEU	6.3387
ClinicalBERT	BLEU	0.00289

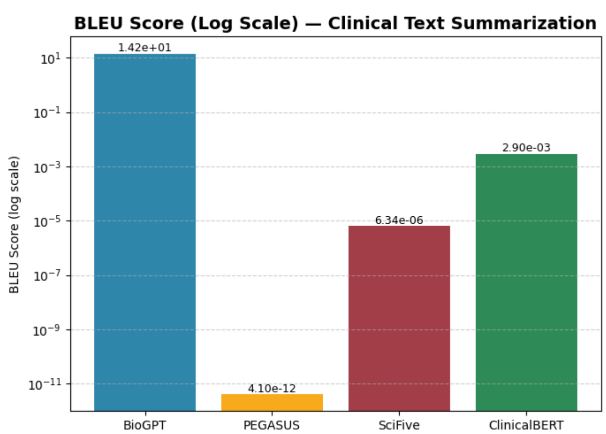


Figure 3: LOG Scale of all 4 models to show magnitude of BLEU score metrics

4.3 BertScore(s)

The BERTScore results reinforced this overall trend. BioGPT recorded the highest precision, recall, and F1 values (all in the 0.98 range) demonstrating a superior ability to preserve the meaning and semantic content of the original text. The other models achieved noticeably lower semantic similarity scores, and although ClinicalBERT performed reasonably well for an extractive approach, it still fell short of BioGPT's level of semantic alignment.

Table 3: **Table 4: BERTScore evaluation metrics (Precision, Recall, and F1-score) for all summarization models.**

Model	Metric	Precision	Recall	F1-Score
BioGPT	BERT Score	0.9864	0.9846	0.9855
PEGASUS	BERT Score	0.8038	0.7672	0.7851
SciFive	BERT Score	0.8712	0.7907	0.8290
ClinicalBERT	BERT Score	0.8557	0.8117	0.8332

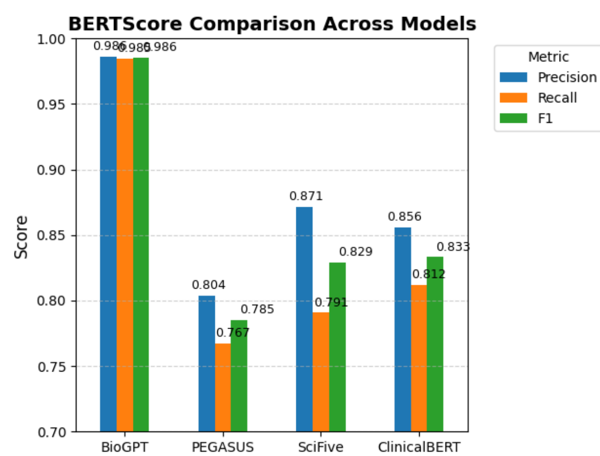


Figure 4: BERT-Score across all four models showing the precision, recall and F1-score

4.4 Results Overall

The study finds that BioGPT was the strongest summarization model among those evaluated. Its performance across ROUGE, BLEU, and BERTScore metrics showed consistent superiority in both lexical accuracy and semantic preservation. These results suggest that BioGPT provided the most faithful, coherent, and meaningful summaries relative to the reference material.

5 Discussion / Conclusion

5.1 Discussion

The objective of the paper was to determine which NLP summarization technique is useful in summarizing patient clinical text records. Both abstractive and extractive summarization were analyzed using 5 different models highlighted above. The significance of the result provides a detailed framework on which summarization model is preferred when analyzing huge clinical text data and its advantage over other models. From the evaluation metrics conducted, BioGPT outperformed the other 3 models (SciFive, PEGASUS and ClinicalBERT) on all metrics. A tentative reason to why BioGPT had the highest F1-score among the three abstractive models could be associated to its training and domain. BioGPT model was made specifically for biomedical literature while the two (PEGASUS and SciFive) are considered to be general abstractive models. BioGPT also outperformed Clincial BERT (an extractive summarization model). Extractive summarization model are not useful in generating human like sentences, often regurgitating the exact sentences it was trained on which inherently defeats the purpose of a shorter concise summary. A few limitations and setbacks were encountered during the analysis of the clinical text dataset. A common setback encountered was the size of the data that was meant to be analyzed. A disadvantage of BioGPT and other models such as PEGASUS and SciFive is the lack of ability to generate large amounts of token which makes it burdensome in comparison to other abstractive models like Longformer (an encoder/decoder model). To compensate for this, three chunks of 450 words were created, another option to circumvent this setback was to utilize the first 6 clinical cases with the intention of scaling this to other similar cases. Possible future work is to examine the role of model transformers and its effect on each summarization model in producing human like readable sentences for abstractive summarization.

5.2 Conclusion

The main goal of the paper was to determine the performance of multiple models in summarizing biomedical data. First, two types of summarization (abstractive and extractive) were highlighted. Multiple models that encompass both type of summarization were used to compare the effectiveness of each summarization technique. An evaluation was conducted to perform analysis on each model. From our result obtained, Bio-GPT outperformed all other models in all three evaluation metrics as seen based on its BERT score and F1-score. We speculated that the reason for this result could be due to the domain and training of the model, which is predominantly trained on biomedical data making it easy to process. Future work should investigate the role of transformers in producing concise human-readable summaries.

6 References

beginitemize

Dahl, S., Bøgsted, M., Sagi, T., and Vesteghem, C. 2025. Performance of natural language processing for information extraction from electronic health records within cancer: Systematic review. *JMIR Medical Informatics*, 13: e68707. https://doi.org/10.2196/68707

Jerfy, A., Selden, O., and Balkrishnan, R. 2024. The growing impact of natural language processing in healthcare and public health. *Inquiry: A Journal of Medical Care Organization, Provision and Financing*, 61: 469580241290095. https://doi.org/10.1177/00469580241290095

Neupane, S., Mitra, S., Mittal, S., Golilarz, N. A., Rahimi, S., Amirlatifi, A., and Gaur, M. 2024. MedInsight: A multi-source context augmentation framework for generating patient-centric medical responses using large language models. *ACM Transactions on Computing for Healthcare*. (Advance online publication)

Van Veen, D., Van Uden, C., Blankemeier, L., Delbrouck, J.-B., Aali, A., Bluethgen, C., Pareek, A., Polacin, M., Pontes Reis, E., Seehofnerová, A., Rohatgi, N., Hosamani, P., Collins, W., Ahuja, N., Langlotz, C. P., Hom, J., Gatidis, S., Pauly, J., and Chaudhari, A. S. 2023. Clinical text summarization: Adapting large language models can outperform human experts. *Research Square*. https://doi.org/10.21203/rs.3.rs-3483777/v1

endthebibliography

enditemize